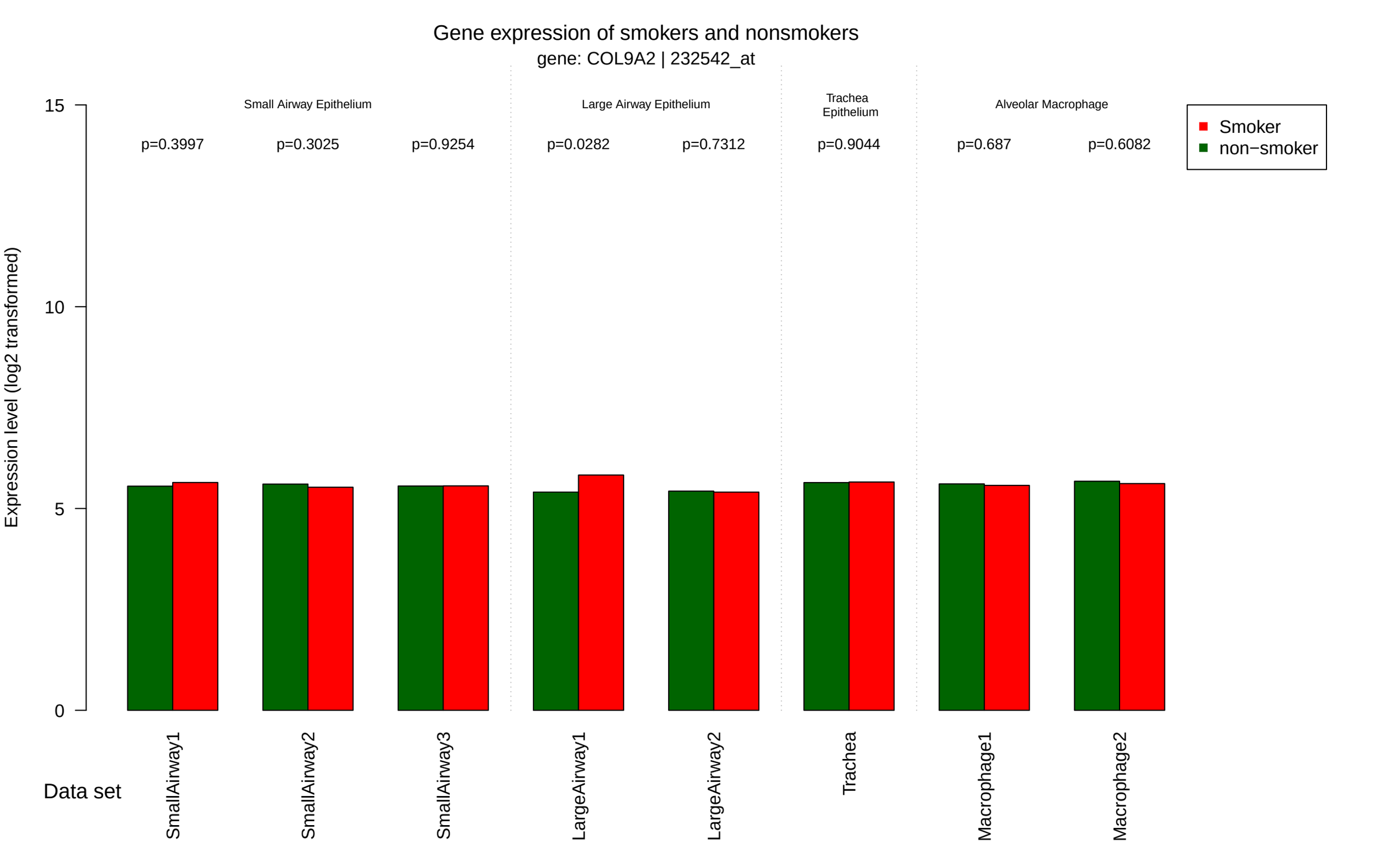
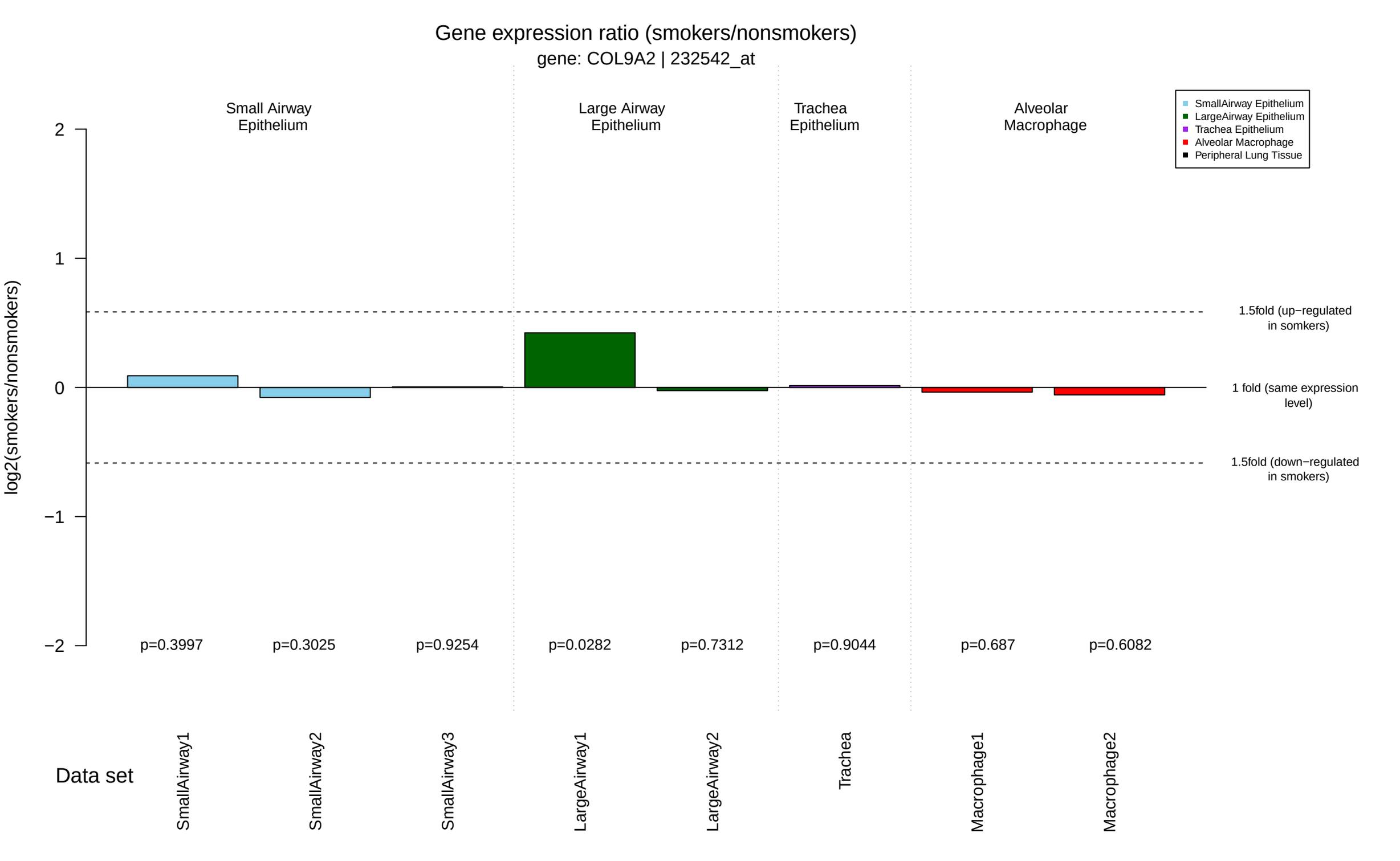
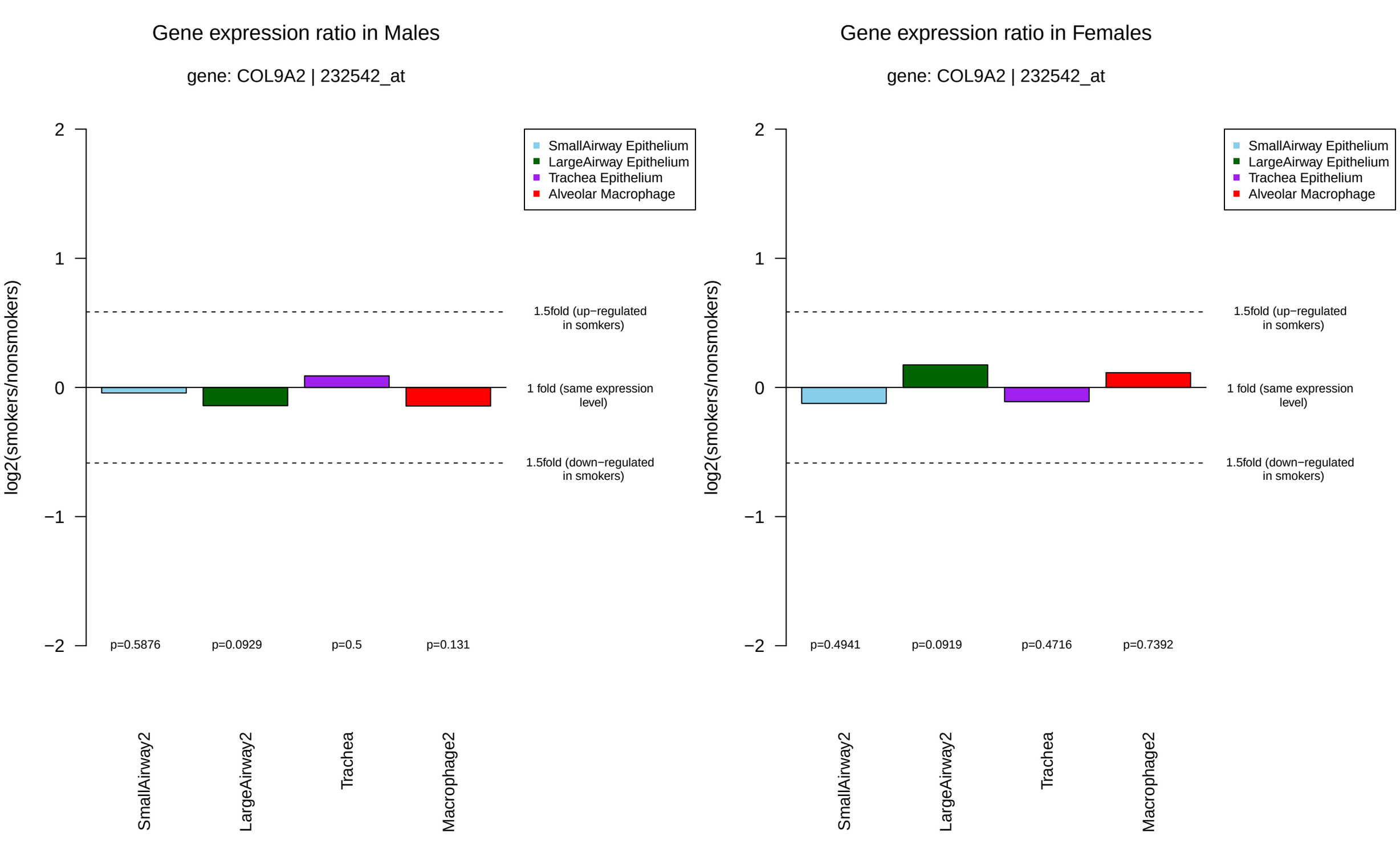
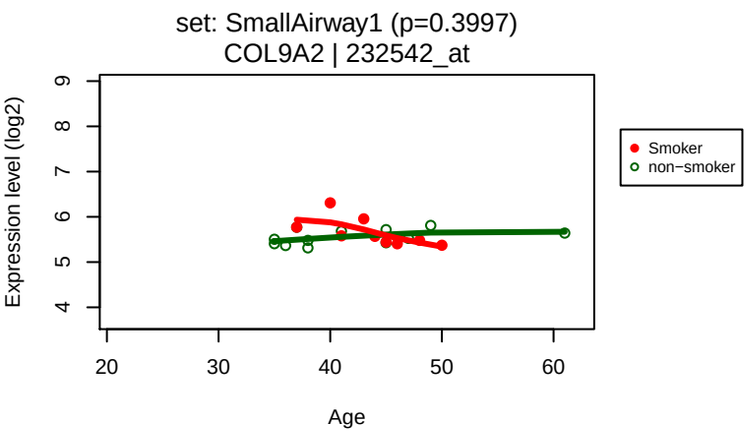
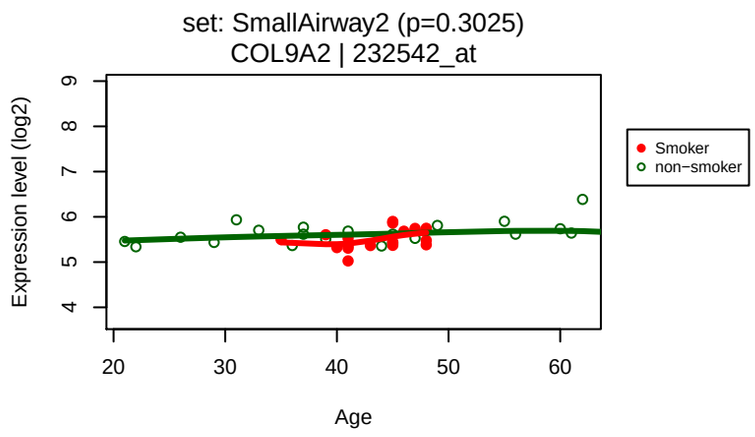
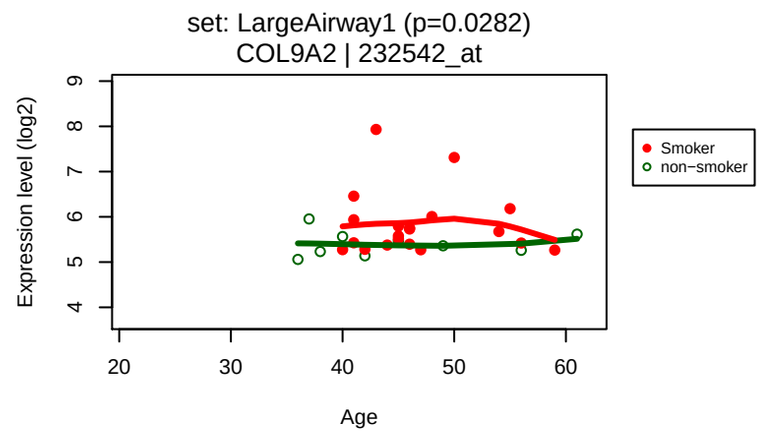

Gene expression of smokers and nonsmokers
gene: COL9A2 | 232542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3997
p=0.3025
p=0.9254
p=0.0282
p=0.7312
p=0.9044
p=0.687
p=0.6082
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

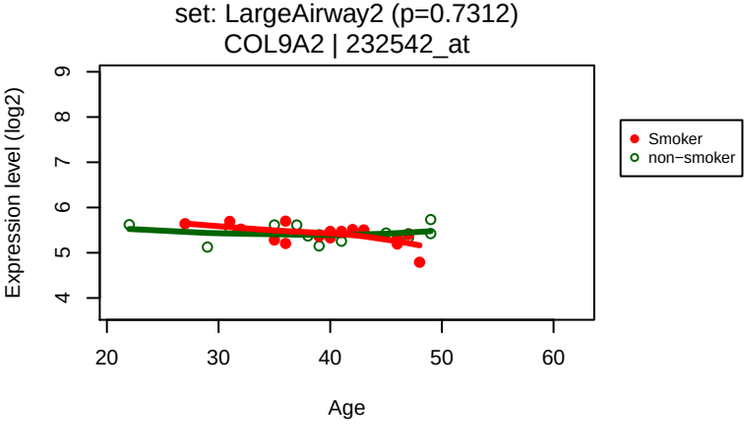

Gene expression ratio (smokers/nonsmokers)
gene: COL9A2 | 232542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3997
p=0.3025
p=0.9254
p=0.0282
p=0.7312
p=0.9044
p=0.687
p=0.6082
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

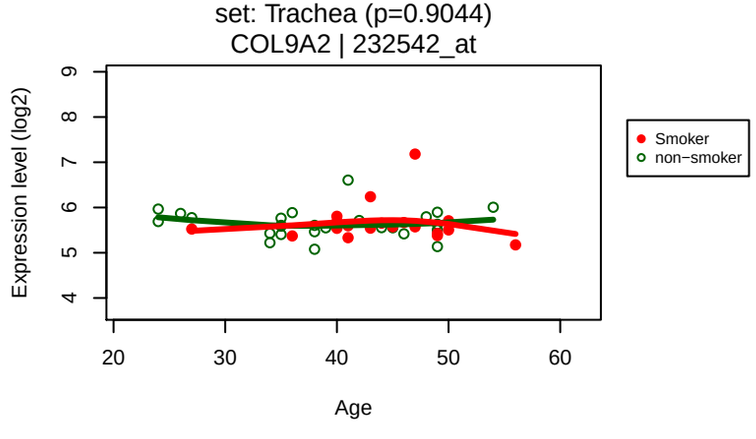

Gene expression ratio in Males
gene: COL9A2 | 232542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5876
p=0.0929
p=0.5
p=0.131
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL9A2 | 232542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4941
p=0.0919
p=0.4716
p=0.7392
SmallAirway2
LargeAirway2
Trachea
Macrophage2

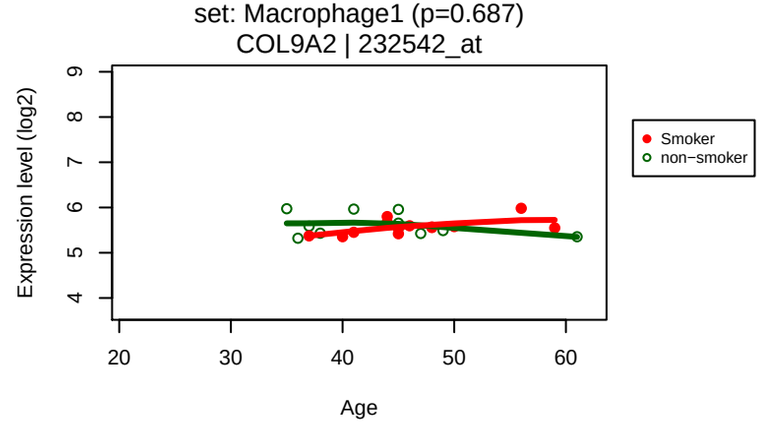

set: SmallAirway1 (p=0.3997)
COL9A2 | 232542_at
Expression level (log2)
Age
Smoker
non−smoker

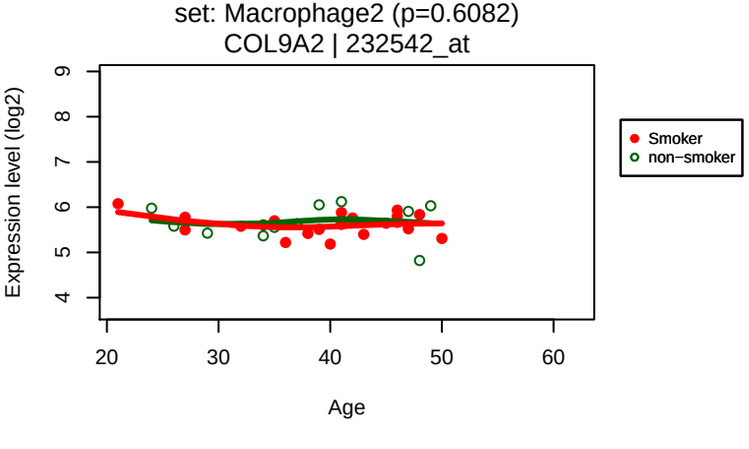

set: SmallAirway2 (p=0.3025)
COL9A2 | 232542_at
Expression level (log2)
Age
Smoker
non−smoker

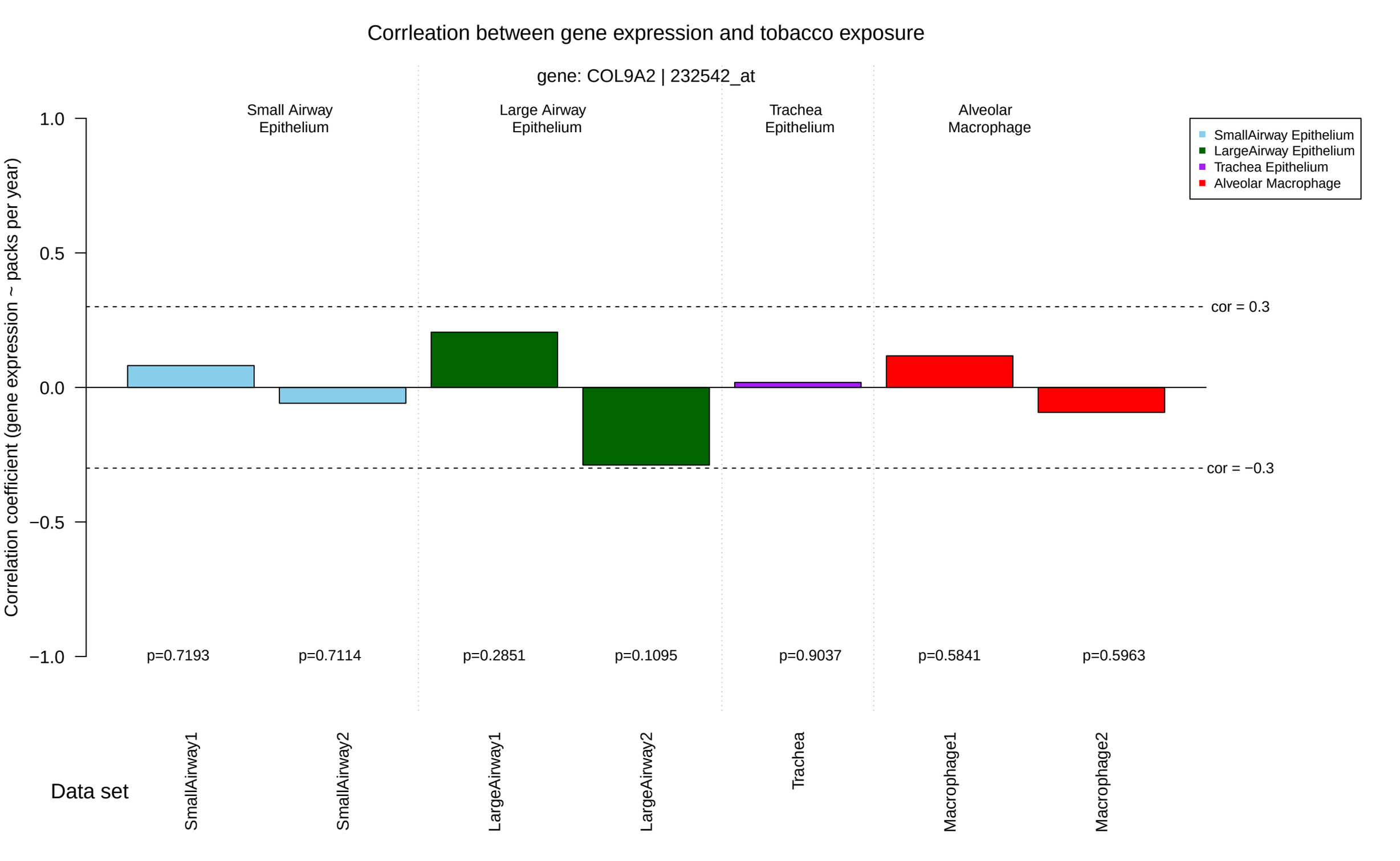

set: LargeAirway1 (p=0.0282)
COL9A2 | 232542_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7312)
COL9A2 | 232542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9044)
COL9A2 | 232542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.687)
COL9A2 | 232542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6082)
COL9A2 | 232542_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL9A2 | 232542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7193
p=0.7114
p=0.2851
p=0.1095
p=0.9037
p=0.5841
p=0.5963
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2